### 7.2.2 Properties of the system:

#### Intensive and extensive properties

Some of the properties of a system depend on its mass or size whereas other properties do not depend on its mass or size. Based on this, the properties of a system are grouped as ***extensive*** property and ***intensive*** property.

#### Extensive properties:

The property that depends on the mass or the size of the system is called an extensive property.

**Examples:** Volume, Number of moles, Mass, Internal energy, etc.,

#### Intensive properties:

The property that is independent of the mass or the size of the system is called an intensive property.

**Examples :** Refractive index, Surface tension, density, temperature, Boiling point, Freezing point, molar volume, etc.,

**Table: 7.1 Typical List of Extensive and Intensive properties**

| Extensive properties | Intensive properties |
|---|---|
| volume, mass, amount of substance (mole), energy, enthalpy, entropy, free energy, heat capacity | molar volume, density, molar mass, molarity, mole fraction, molality, specific heat capacity |

### 7.2.3 Thermodynamic processes

The method of operation which can bring about a change in the system is called thermodynamic process. Heating, cooling, expansion, compression, fusion, vaporization etc., are some examples of a thermodynamic process.

#### Types of processes:

A thermodynamic process can be carried out in different ways and under different conditions. The processes can be classified as follows:

#### Reversible process:

The process in which the system and surrounding can be restored to the initial state from the final state without producing any changes in the thermodynamic properties of the universe is called a reversible process. There are two important conditions for the reversible process to occur. Firstly, the process should occur infinitesimally slowly and secondly throughout the process, the system and surroundings must be in equilibrium with each other.

#### Irreversible Process:

The process in which the system and surrounding cannot be restored to the initial state from the final state is called an irreversible process. All the processes occurring in nature are irreversible processes. During the irreversible process the system and surroundings are not in equilibrium with each other.



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